SpanDiff technical report

Summary

SpanDiff scores token-level semantic drift on Swiss federal pages in German, French, and Italian. It is a representation model, not a prompt. The submitted scorer is hidden state 15 of swiss-ai/Apertus-v1.5-8B using the pre-registered paper-order subword-first formulation (docs/preregistration.md): each content subword is scored by clamp(1 - max cosine to the other side, 0, 1), and subword scores are averaged inside the whitespace token they overlap most. No tokens are generated. Apertus 70B was not run.

The official development script, with --split dev, printed Spearman de 0,309 ; fr 0,199 ; it 0,333. The unweighted mean of those three printed numbers is 0.2803. The official script does not print that mean. make run prints it after the table. Kendall's Tau-b printed de 0,249 ; fr 0,161 ; it 0,266. The stdout is track_2a/docs/official_dev_stdout_subword_l15.txt.

Setup	de	fr	it	mean
Paper DiffAlign on this 8B, final layer, clamped	0.053	0.056	0.106	0.0717
Hidden state 8 ablation (word mean-pool)	0.245	0.165	0.282	0.2307
Pre-registered (a): mean of hidden states 12-17 (word mean-pool)	0.285	0.183	0.316	0.2613
Hidden state 15 ablation (predictions/l15, word mean-pool)	0.296	0.191	0.324	0.2703
Pre-registered (c): hidden states 12-17 + subword-first	0.298	0.192	0.325	0.2717
Hidden state 15, submitted (paper-order subword-first)	0.309	0.199	0.333	0.2803

The UZH challenge asks for a SwissGov-paper baseline on this same Apertus model. That baseline is the clamped final-layer row, mean 0.0717. The submitted mean is 0.2803 (and 0.2703 for the word-mean-pool hidden-state-15 ablation in predictions/l15). The mean column is that unweighted average, shown to 4 decimals. The rows use the printed coefficients. Decimal commas in the stdout are 0,309 and so on. Hidden state 15 word mean-pool is docs/official_dev_stdout_L15.txt. Pre-registered variants (a), (b), and (c) are docs/official_dev_stdout_l12_17.txt, docs/official_dev_stdout_subword_l15.txt, and docs/official_dev_stdout_combined.txt. Hidden state 8 is docs/official_dev_stdout.txt. The final layer is docs/official_dev_stdout_last.txt. All grades used --split dev on 168 lines per language.

Method

Each side of a page is encoded alone. The model is swiss-ai/Apertus-v1.5-8B. Index 0 is the embedding. The submitted state is index 15, the residual stream after block 15. Index 32, after the final norm, is the final-layer baseline, not the submission.

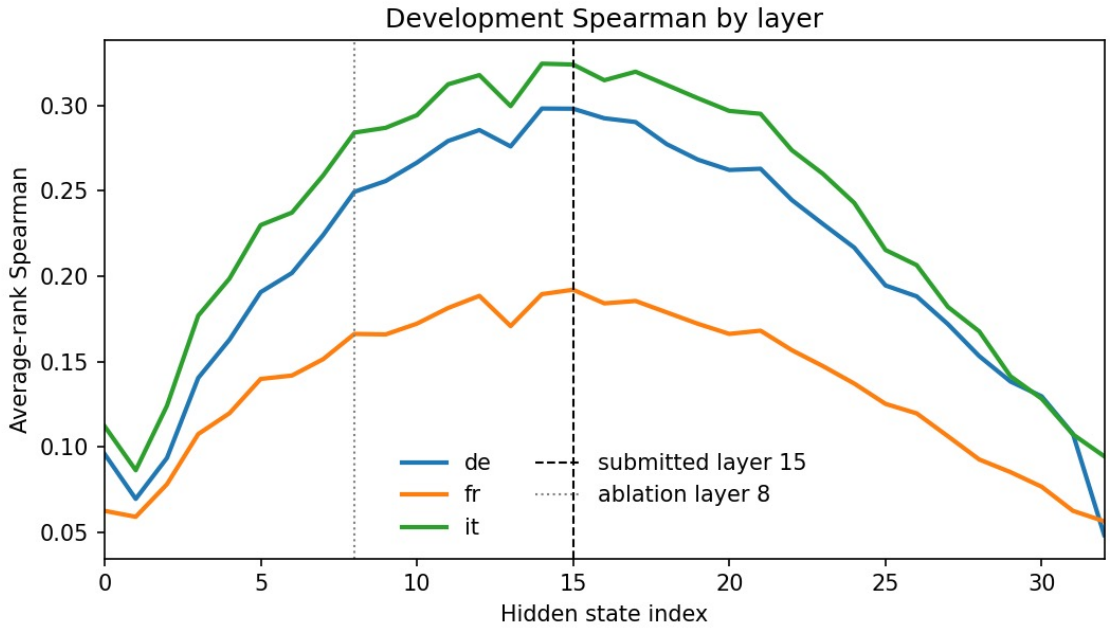
A window holds at most 512 tokenizer tokens, including the leading beginning-of-sequence token. The development pass is 1619 such windows and 564,349 tokenizer tokens. Each of those windows has exactly one leading BOS. That token has an empty character offset and is not pooled. EOS, PAD, UNK, and multimodal special counts on that tokenizer-only census were 0. The census is docs/special_tokens.txt. It did not load weights.

Each subword is assigned to the whitespace token it overlaps most. An equal overlap stays on the earlier token. In the submitted subword-first method (pre-registered variant (b)), after the windows of one side are stitched, each content subword is scored against all content subwords on the other side by clamp(1 - max cosine, 0, 1), and the subword scores inside each whitespace token are averaged and rounded to 6 decimals. A whitespace token with no subword scores 1. In the word-mean-pool ablation (predictions/l15), subword hidden states are averaged into the whitespace token first and then scored by clamp(1 - max cosine, 0, 1).

The paper's DiffAlign takes the final hidden state and does not clamp, so a negative cosine can score above 1. SpanDiff clamps. The final-layer row in the table is the clamped version of that recipe on this 8B model. An unclamped rescore of the saved final-layer states was not produced.

Encoder lines contain only id, text_a, text_b, labels_a, and labels_b. Every label is in [0, 1]. The writer rejects -1 and NaN. The label count equals the whitespace-token count. Generated tokens are 0, so no output filter is applied.

Layer curve



The figure plots the saved average-rank Spearman for every hidden state, separately for German, French, and Italian. The source table is docs/layer_map_by_lang_avg.txt. It uses the same token counts as the frozen-score tables (German 108,945, French 124,262, Italian 118,735) and drops gold labels of -1. It is not a second run of the official script. Do not treat a point on the curve as the printed contest score.

On that map, hidden state 8 reads 0.2495, 0.1662, and 0.2842. Hidden state 15 reads 0.2982, 0.1921, and 0.3241. German peaks at layer 14 (0.2983), a hair above layer 15. French peaks at layer 15. Italian peaks at layer 14 (0.3246), again a hair above layer 15. The final state, index 32, is 0.0484, 0.0564, and 0.0945. German is the low end of that curve. A pooled ordinal Spearman across all three languages, in docs/layer_map_dev.txt, peaks at layer 14 (0.1408), with layer 15 at 0.1403, layer 8 at 0.1281, and the last block at -0.0358. That pooled number is one Spearman on 351,942 tokens. It is not the mean of three language scores.

The official prints sit next to this map, not on top of it. Submitted subword-first hidden state 15 printed 0,309 ; 0,199 ; 0,333 (mean 0.2803). Word-mean-pool hidden state 15 printed 0,296 ; 0,191 ; 0,324 (mean 0.2703). Hidden state 8 printed 0,245 ; 0,165 ; 0,282 (mean 0.2307). The final layer printed 0,053 ; 0,056 ; 0,106 (mean 0.0717). The shape agrees: the middle of the network carries the signal, and the final state does not.

docs/token_heatmap.png and docs/demo.html show development pairs (admin_de_105, admin_fr_141, admin_it_101) colored by the clamped difference alongside gold labels.

Layer selection and pre-registered representation study

The organizers publish a development train split of 134 documents per language and a development validation split of 34, at gold_labels/dev/train and gold_labels/dev/val in ZurichNLP/SwissGov-RSD. Those files are not copied into this repository. Only the id field was read. Together, train and validation are the 168 development documents, and they share no id. The per-document Spearmans were then taken from docs/layer_map_dev.jsonl, which stores one ordinal Spearman per hidden state and does

not store gold labels.

A document whose per-document Spearman is not finite was omitted: none of 134 German train pages, 1 of 34 German val, 2 of 134 French train, 1 of 34 French val, 1 of 134 Italian train, and 1 of 34 Italian val. The table is the unweighted mean of the remaining per-document Spearmans. It is not the official contest formula.

Slice	docs used	best layer	layer 8	layer 15	15 minus 8
de train	134/134	15	0.1623	0.1899	+0.0276
de val	33/34	15	0.1636	0.2003	+0.0367
fr train	132/134	15	0.0430	0.0798	+0.0367
fr val	33/34	15	0.0461	0.0801	+0.0340
it train	133/134	14	0.1509	0.1903	+0.0394
it val	33/34	15	0.1796	0.2066	+0.0270

Italian train is the exception. Its best layer is 14, at 0.1913, and layer 15 is 0.1903. An n-weighted mean of the same per-document values picks the same best layer in every row, including Italian train at layer 14. Layer 15 beats layer 8 by about 0.03 on every slice.

Before running further representation changes, three variants on swiss-ai/Apertus-v1.5-8B were pre-registered in docs/preregistration.md with a strict gate: beat word-mean-pool L15 on dev/train (134/lang), confirm on dev/val (34/lang), and beat full-dev official grade by at least +0.0050. On dev/train and dev/val token-pooled Spearman, variant (a) mean_l12_17 scored 0.2641 / 0.2457 (official full-dev 0.2613), variant (c) combined scored 0.2750 / 0.2529 (official full-dev 0.2717), and variant (b) subword_first_l15 scored 0.2843 / 0.2592 vs word-mean-pool L15 at 0.2734 / 0.2527, and achieved 0.2803 on the official full-dev grade (+0.0100 over 0.2703). Variant (b) therefore became predictions/dev. Word-mean-pool L15 remains in predictions/l15, and hidden state 8 remains in predictions/ablation_l8.

Use of Apertus

One checkpoint: swiss-ai/Apertus-v1.5-8B. No second embedding model and no closed API. The development forward pass read 564,349 tokens and wrote 0 generated tokens. With the weights already cached, layer-15 subword-first scoring took 549.8 seconds on one Apple GPU. A cold load of the 8B weights needs about 16 GB. Apertus 70B was not run. It does not fit in 48 GB on this machine. The Getting Started introduction says builders work with both sizes. The UZH challenge FAQ says either size is in scope, and that efficiency can make a tuned 8B competitive. This submission is the 8B.

Test predictions, 56 documents per language, and full-set predictions, 224 documents per language, use the same frozen hidden-state-15 subword-first method. The test forward pass was 207,075 tokens and 261.8 seconds. Development scores are our development evaluation. Those test and full-set files are included for the organizers to score. We never graded the test split. No test Spearman is claimed.

Data

Development inputs are the public SwissGov-RSD development pages, 168 documents per language. Test predictions are 56 documents per language. Full-set predictions are 224 documents per language, in id order. Development scores are our development evaluation. The test and full-set predictions are included for the organizers to score. We never graded the test split. This repository does not contain that gold. The SwissGov text stays CC-BY-4.0 and is not relicensed.

Evaluation

The summary table is the evaluation that should be quoted. It has six official development runs (submitted subword-first L15, pre-registered (c), word-mean-pool L15, pre-registered (a), L8, and clamped final layer). Layer-map and pre-registration numbers explain why subword-first hidden state 15

was adopted.

Limitations

States are causal, so a token sees only its left context. An empty subword span scores 1, which inflates a few punctuation tokens. The final layer raises aligned tokens and different tokens together, so a last-layer DiffAlign score looks large and ranks poorly. Italian train, on the per-document map, prefers layer 14 by 0.001. The unclamped paper formula was not rerun. 70B was not run. No test Spearman is reported. We never graded the test split.

Reproducibility

The public tree is https://github.com/moscraciunxxx/SpanDiff-HackApertus. Judges run make run. That builds the Docker image and grades predictions/dev with --split dev. It does not load the model. make run-local is that grade when Docker is not installed. LLM_NAME, LLM_BASE_URL, and LLM_API_KEY are not the scorer. make predict writes a new deterministic layer-15 development run under predictions/rerun. It needs the Apertus weights and a Hugging Face token only if those weights are not cached. The Docker image grades the same committed development files. An earlier image was built and graded under podman on linux/amd64. Docker is not installed on this Mac, so the current image has not been rebuilt here.

Next steps

Development scores are our development evaluation. Test predictions, 56 documents per language, and full-set predictions, 224 documents per language, use the frozen hidden-state-15 subword-first method and are included for the organizers to score. We never graded the test split. make run grades the development files only. A 70B run is future work on a larger machine.

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